

Ten New Age Technologies Built on Open Source

Let's take a quick look at ten new age technologies that are fuelled by or almost fully based on open source.

1



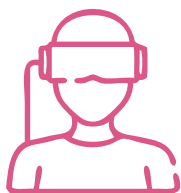
AI or artificial intelligence

In simple words, AI refers to the intellectual intelligence of smart machines in areas like problem-solving, taking decisions, human-like comprehension and perception. As opposed to their conventional counterparts that just follow instructions and execute as

programmed, AI-powered machines can not only take instance-based decisions but also gain more efficiency and capabilities over a period of time.

AI includes reactive AI, and machines with limited memory, mind theory and self-awareness. Though work on the concept began in the 1960s, it was in 2006 that the term gained popularity. Today AI is used in one way or the other across different sectors like banking and finance, retail, government and healthcare, among others.

5



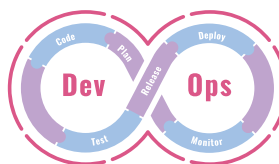
AR/VR or augmented reality/virtual reality

AR or augmented reality refers to the interactive experience brought about by the strategic use of digital audio/video elements. VR or virtual reality refers to simulating real-life scenes so sophisticatedly that

a user feels fully immersed in the surroundings. It is achieved through advanced computer modelling, 3D visuals and a finely simulated environment. There are three broad types of VR — non-immersive, semi-immersive and fully-immersive. While AR works in a real-world setting and can be experienced on a mobile device, VR is fully virtual in nature and you will require a headset to experience it. While VR works on an entirely virtual model, AR enhances both spheres — real and virtual.

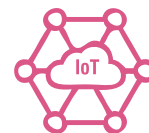
2

DevOps or development and operations



DevOps refers to different tools and practices that combine software development and operations — hence, the name DevOps. It works in the areas of software development and IT operations by automating production and deployment processes between them to significantly speed up tasks, eliminate execution discrepancies and delays, and thus enhance the organisation's IT productivity. DevOps works throughout the entire SDLC (software development life cycle). As compared to the conventional infrastructure management process and development methods, DevOps can help improve products at a quick speed due to its robust architecture.

3



IoT or Internet of Things

IoT or Internet of Things refers to a system equipped with sophisticated software, sensors and various high-end technologies. IoT seamlessly connects with a massive number of computing devices, machines and digital devices over communication/internet connections across the globe. It enables seamless data exchange between those devices. Open source is the vital enabler of IoT platforms and processes. In simple terms, IoT creates a communication system between different devices to boost their performance and applications.

4



Cyber security methods

Cyber security has become a huge challenge in the modern data-enabled era. With the help of powerful open source based cyber security methods, the probability, frequency and magnitude of cyber attacks can be remarkably reduced. These methods focus on providing failsafe protection against unauthorised system/network exploitation by threat actors or bots.